Exp 11a: Simulation of Static Routing Configuration Using Cisco Packet Tracer

AIM :

To simulate the static routing configuration using the Cisco Packet Tracer simulator.

ALGORITHM :

1. Start the experiment.
2. Create the Packet Tracer lab with three routers (Router0, Router1, Router2), two switches, four PCs and a server, and assign the IP addresses to the devices.
3. Note down the networks directly connected to each router and the networks that are not available on it.
4. On Router0, create two static routes for network 30.0.0.0/8 (main route via Router1, backup route via Router2), two static routes for the host 30.0.0.100 and two static routes for network 50.0.0.0/8 using the `ip route` command with different administrative distance (AD) values.
5. On Router1, create two static routes for network 10.0.0.0/8 and two for network 40.0.0.0/8. On Router2, create static routes for networks 10.0.0.0/8 and 30.0.0.0/8.
6. Verify with `show ip route static` that only the main routes are added to the routing table.
7. Verify the path of the packets using the `tracert` command from a PC of network 10.0.0.0/8.
8. Simulate the failure of the main route by deleting the link between Router0 and Router1, repeat the test and observe that the backup route is used.
9. Delete a static route using the `no ip route` command and observe the result.
10. Stop the experiment.

PROCEDURE :

Static routes are the routes that are added manually to the router’s routing table; this process is known as static routing. Routers automatically learn their directly connected networks, so routes are required only for the networks that are not available on the router’s interfaces.

Setting up the lab: Create the Packet Tracer lab as shown in Fig 1. Each network has two routes to reach it; one route is configured as the main route and the other as the backup route. If two routes to the same destination are specified, the router adds the route with the lower AD value to the routing table (for example AD 10 for the main route and AD 20 for the backup route). If the main route fails, the router automatically adds the backup route.

Networks of each router:

Router	Networks on local interfaces	Networks on other routers
Router0	10.0.0.0/8, 20.0.0.0/8, 40.0.0.0/8	30.0.0.0/8, 50.0.0.0/8
Router1	20.0.0.0/8, 30.0.0.0/8, 50.0.0.0/8	10.0.0.0/8, 40.0.0.0/8
Router2	40.0.0.0/8, 50.0.0.0/8	10.0.0.0/8, 20.0.0.0/8, 30.0.0.0/8

Router0 configuration: main route for 30.0.0.0/8 via Router1, main route for host 30.0.0.100 via Router2 and main route for 50.0.0.0/8 via Router2; the other routes are the backup routes.

```
Router>enable
Router#configure terminal
Router(config)#ip route 30.0.0.0 255.0.0.0 20.0.0.2 10
Router(config)#ip route 30.0.0.0 255.0.0.0 40.0.0.2 20
Router(config)#ip route 30.0.0.100 255.255.255.255 40.0.0.2 10
Router(config)#ip route 30.0.0.100 255.255.255.255 20.0.0.2 20
Router(config)#ip route 50.0.0.0 255.0.0.0 40.0.0.2 10
Router(config)#ip route 50.0.0.0 255.0.0.0 20.0.0.2 20
Router(config)#exit
Router#show ip route static
```

Router1 configuration: main route for 10.0.0.0/8 and 40.0.0.0/8 via Router0, backup via Router2.

```
Router>enable
Router#configure terminal
Router(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1 10
Router(config)#ip route 10.0.0.0 255.0.0.0 50.0.0.1 20
Router(config)#ip route 40.0.0.0 255.0.0.0 20.0.0.1 10
Router(config)#ip route 40.0.0.0 255.0.0.0 50.0.0.1 20
Router(config)#exit
Router#show ip route static
```

Router2 configuration: static routes for networks 10.0.0.0/8 and 30.0.0.0/8.

```
Router>enable
Router#configure terminal
Router(config)#ip route 10.0.0.0 255.0.0.0 40.0.0.1
Router(config)#ip route 30.0.0.0 255.0.0.0 50.0.0.2
Router(config)#exit
Router#show ip route static
```

Verifying the static routing: The configuration is verified (i) by sending ping requests to a PC of network 30.0.0.0/8 and tracing the path with the `tracert` command from a PC of network 10.0.0.0/8, and (ii) by listing the routing table entries of Router0. The same test is repeated for the host 30.0.0.100. To test the backup route, the failure of the main route is simulated by deleting the link between Router0 and Router1 and the test is repeated.

Deleting a static route: Use `show ip route static` to print all the static routes, note down the route to be deleted and delete it with the `no ip route` command. If a backup route exists, it becomes the main route when the main route is deleted.

```
Router#configure terminal
Router(config)#no ip route 30.0.0.100 255.255.255.255 40.0.0.2
Router(config)#no ip route 30.0.0.100 255.255.255.255 20.0.0.2
Router(config)#exit
```

OUTPUT :

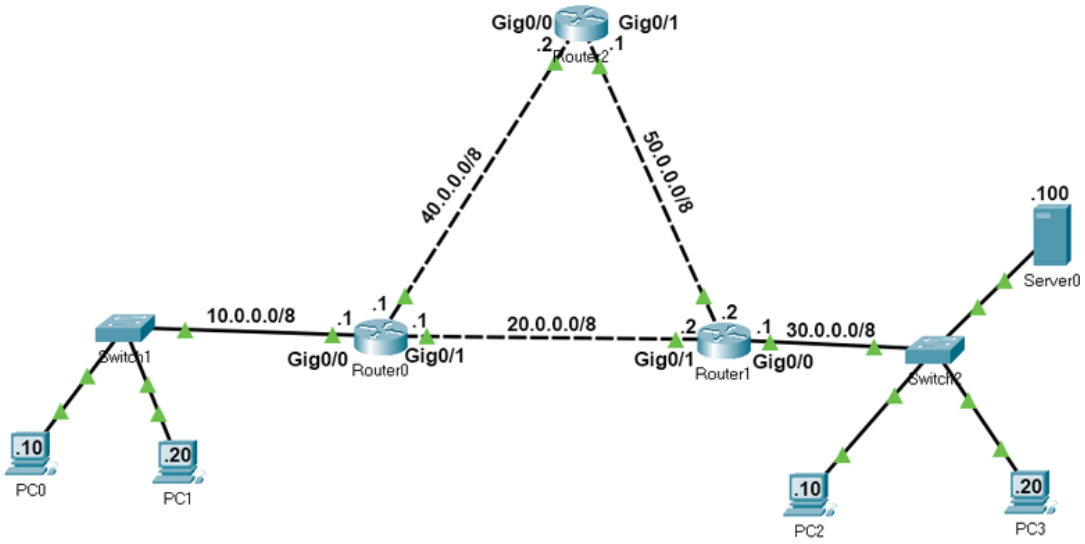


Fig 1. Network topology of the static routing lab

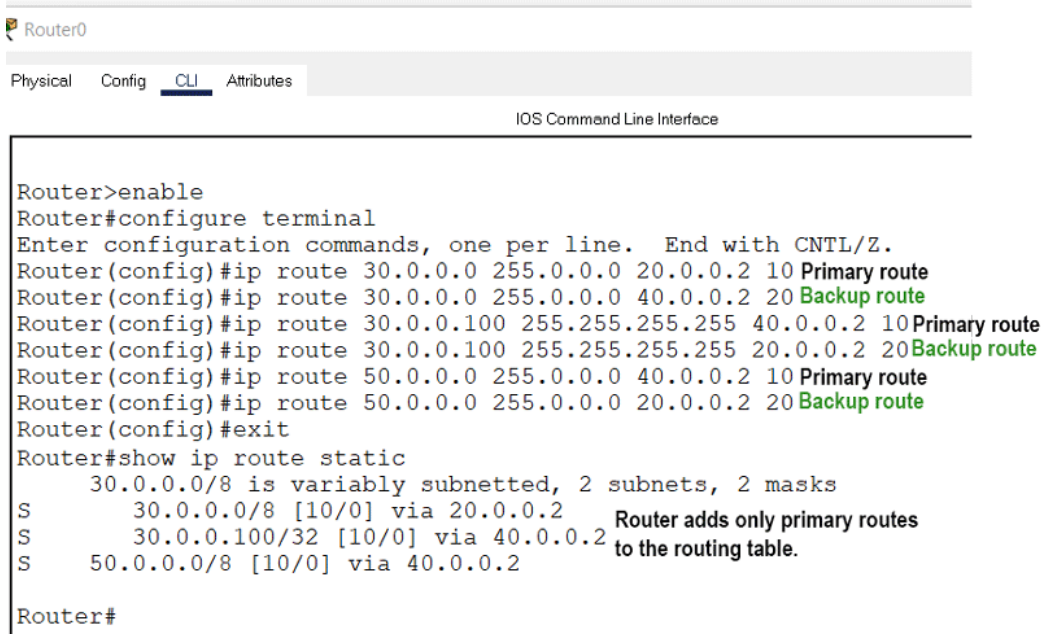


Fig 2. Router0 – static routes configured and routing table (only primary routes are added)

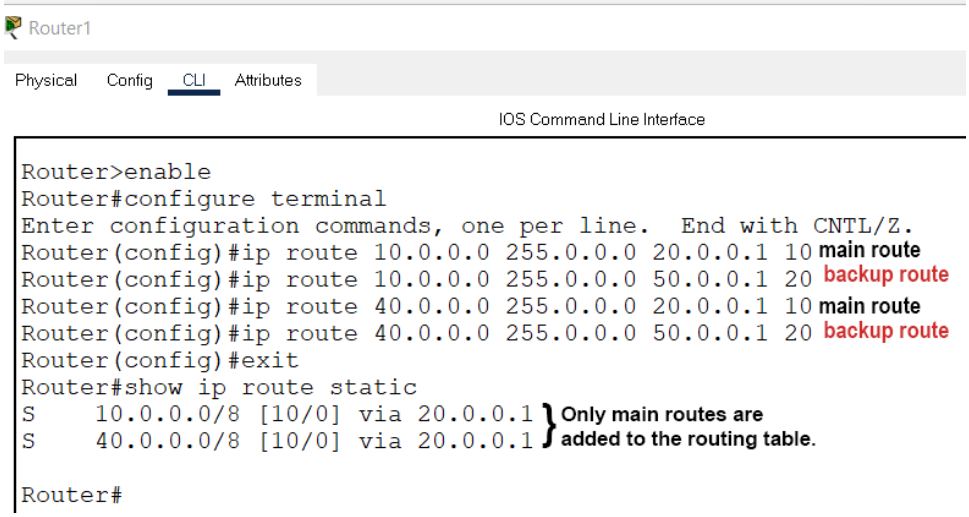


Fig 3. Router1 – static routes configured and routing table (only main routes are added)

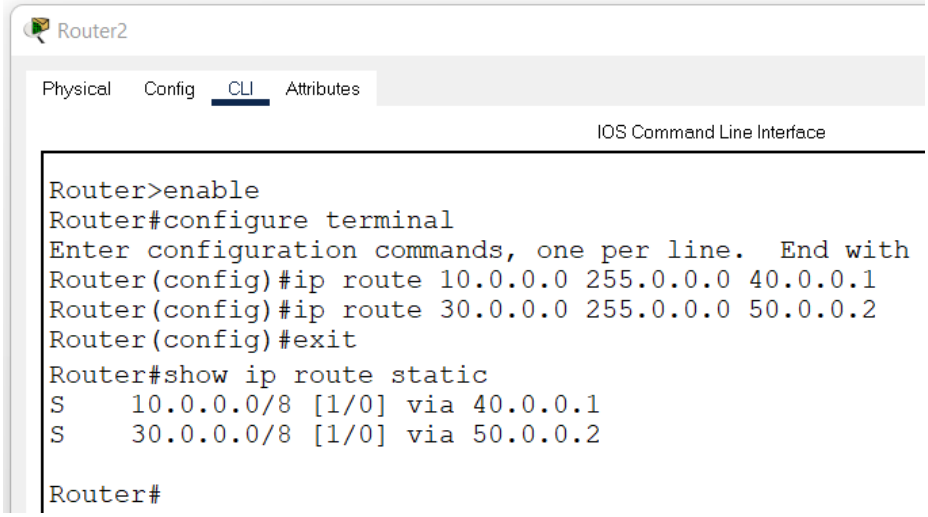


Fig 4. Router2 – static routes configured and routing table

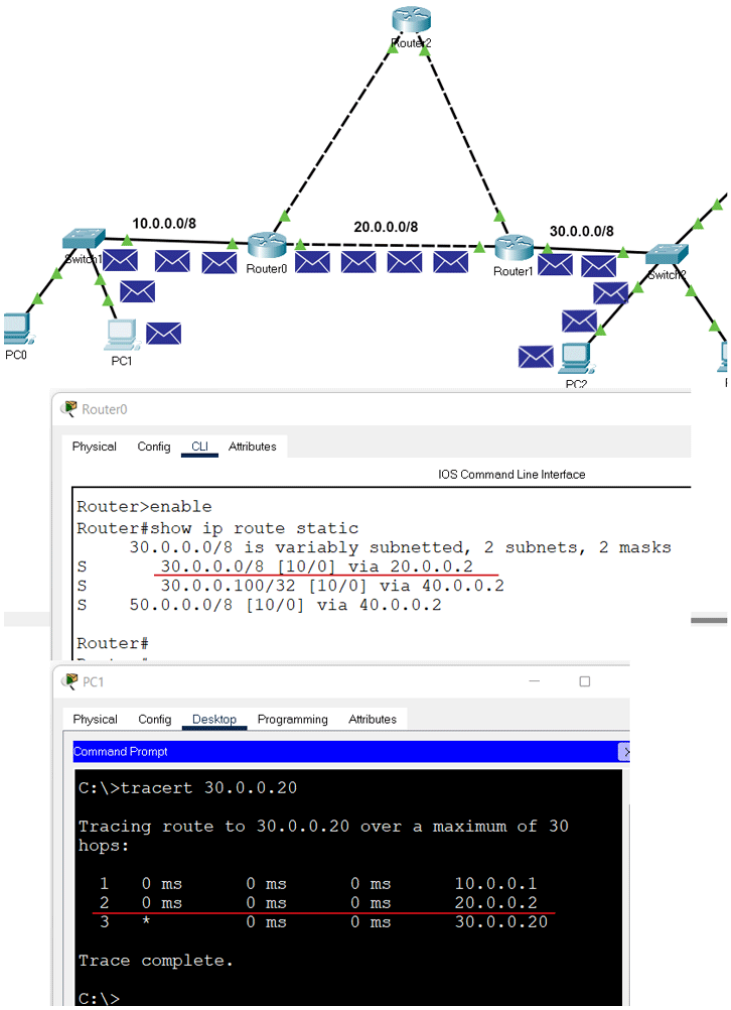


Fig 5. Verification for network 30.0.0.0/8 – packet flow, routing table of Router0 and tracert result

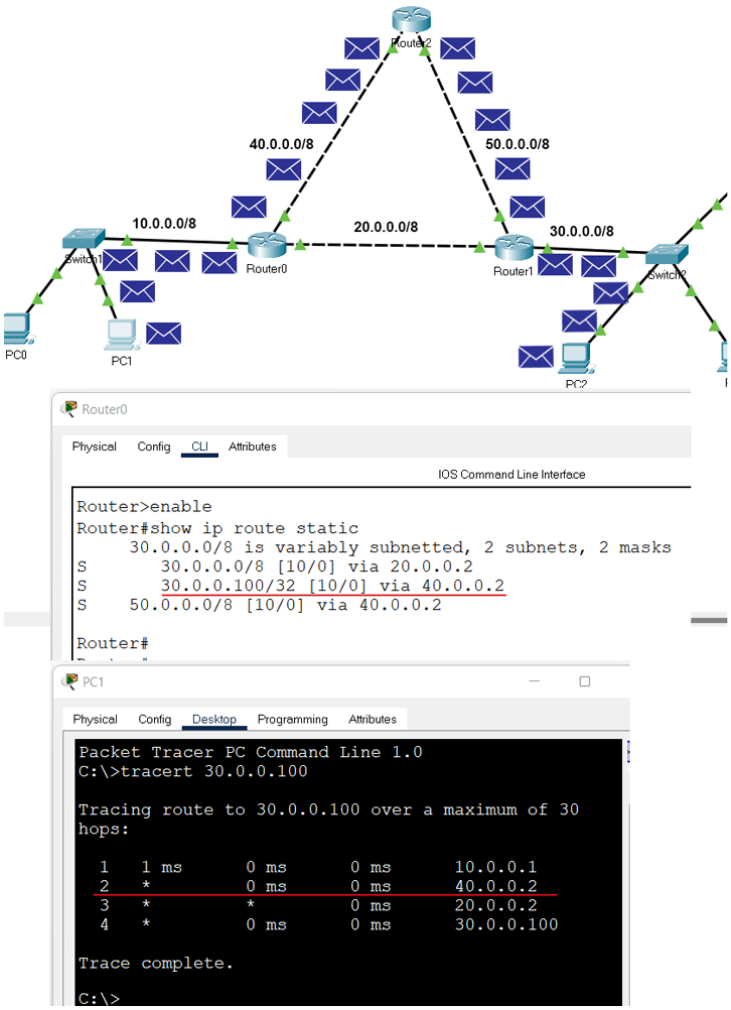


Fig 6. Verification for the host 30.0.0.100 – packet flow, routing table of Router0 and tracert result

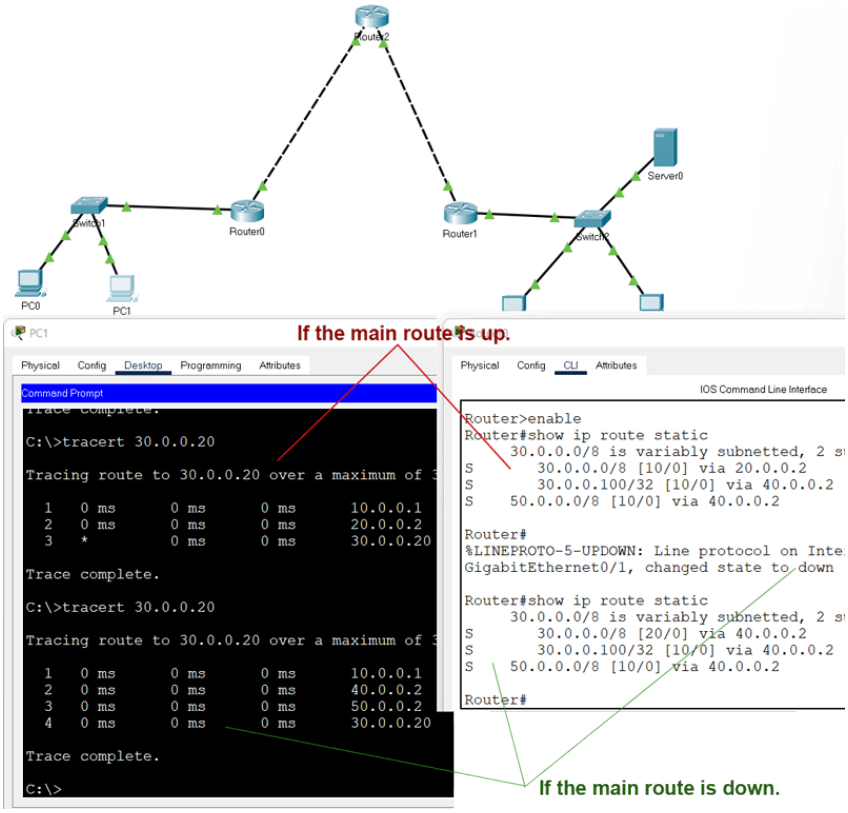


Fig 7. tracert result when the main route is up and when the main route is down (backup route used)

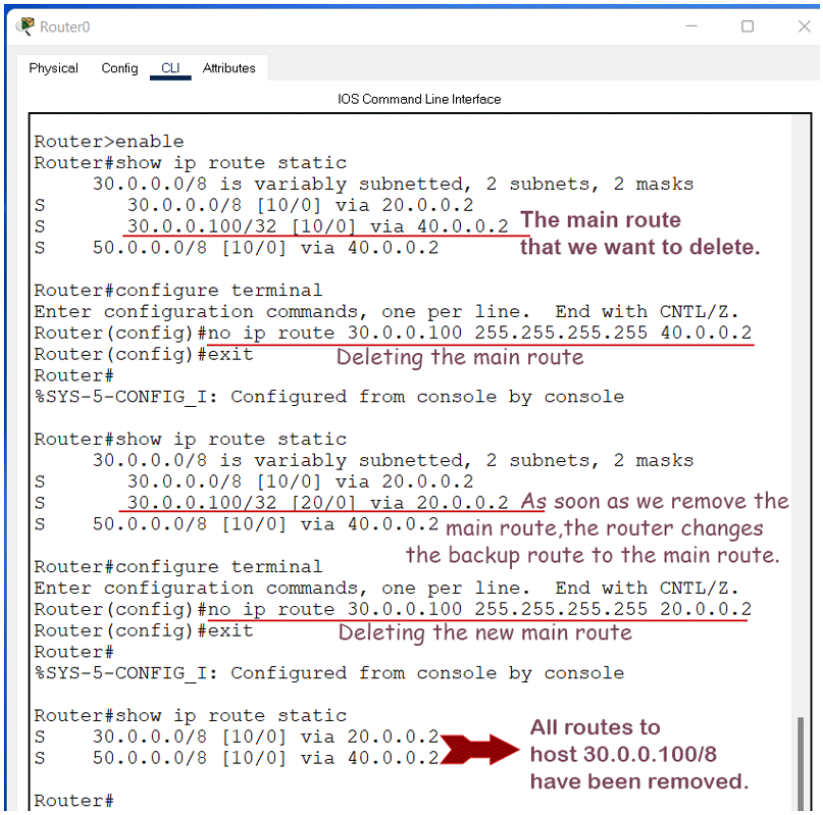


Fig 8. Deleting the static routes of the host 30.0.0.100 (backup route becomes the main route)

RESULT :

Thus, the static routing configuration was simulated using the Cisco Packet Tracer simulator. The main and backup static routes were created and verified, and it was observed that the router uses the backup route when the main route fails.